

Project Progress Report
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Project Name: EchoBreaker

Research Paper Title: EchoBreaker: How People Form and Update Beliefs About Public Opinion During Social Media Browsing

(I will continue working on this project after 6156 with plans to write a research paper. This is the tentative title of that paper.)

Overview:

(Note: I actually started writing my research paper and already wrote part of the initial “Overview” or “Introduction” section. I have taken parts of that and put it into this “Overview” here)

Over the past couple of decades, social media platforms have increasingly become a major source of content consumption, and for some users, the dominant source. As a result, social media plays a major role in how people perceive how their fellow citizens think. Imagine a user browsing through X, formally known as Twitter. As they scroll, the user sees a stream of opinion statements and factual claims, each accompanied by contextual data that shapes how the post lands. This contextual information includes engagement metrics (likes, views, reposts, reply counts), the author of the post, the tone of the most engaged replies, the general sentiment of replies, whether other independent posts share the viewpoint, and whether the post was viewed as a result of an algorithm, among many others. Through this process, users construct impressions of public opinion, which research has shown to be different than actual public opinion, most often measured by polling data. The difference between perceived and actual public opinion is called the perception gap.

Most users are intellectually aware that what appears on social media is often not representative of the sentiments of the general public. They know that the loudest voices usually win, and that algorithmic feeds are curated to optimize for engagement, which often means promoting posts designed to elicit the strongest emotional responses. A motivation behind this paper, therefore, is not to study whether people mistake social media sentiment for general public sentiment, but to understand what and how contextual information influences a users' intuitive estimates of public opinion, despite their intellectual understanding that social media is not a poll. A useful analogy is the Müller-Lyer illusion, a well-studied optical illusion in which lines of identical length appear to be different. Human beings intellectually know that the lines are of identical length; after all, they can literally measure it with a ruler. However, even after measurement, people continue to perceive one as longer than the others and act accordingly.

On its own, perception gaps might seem inconsequential; however, these public opinion distortions do matter because they can influence actual human behavior. For example, perception gaps can produce pluralistic ignorance, a well-studied phenomenon whose core idea can be defined by: "No one believes, but everyone believes that everybody else believes." In essence, social media users might believe their opinion is a minority, even if it is the majority, due to content on a platform's feed. This causes them to be silent and the distorted public sentiment is reinforced for everyone else. What may follow include alienation from society, ill-placed contempt for fellow citizens, erosion of trust in various institutions, unreasonable distrust for "the other side," and radicalization to create a sense of belonging, among many others. On the other hand, false consensus effect, a tendency to overestimate the public support for one's own views, can also have real-world behavioral implications. Especially when people with fringe views have false consensus (often via social media activity), they can become emboldened to further engage in activities that perpetuate their views and ideas.

There has been ample literature studying which topics are most susceptible to perception gaps and how wide they are; this is usually quantified by asking research participants to estimate public opinion on some topic (e.g. politics, technology, lifestyle) and then comparing it to actual polling data. In addition, there has been a substantial amount of research on misinformation within social media and how to combat it, often with various fact-checking tools. While both bodies of work are greatly valuable, this paper asks two different, but related questions: first, how do perception gaps form in real-time within the specific context of a specific social media post? How much do engagement metrics and content of replies influence these gaps versus one's own pre-existing opinions? Second, what happens when users are confronted with potential contradictory data exactly at the gap formation stage? Do they update their beliefs? Are they reflective or combative?

To address these questions, we introduce EchoBreaker, a web browser extension that lets users select specific social media posts (tweets, reddit posts/comments, youtube comments, etc.) and shows you a simulated public sentiment breakdown using LLM synthetic surveys. While to our best knowledge, EchoBreaker is the first tool of its kind, it is not the main contribution of this paper. Rather, EchoBreaker is a research artifact to enable our study.

Research Questions:

RQ1 (Formation): What social media contextual data and what user characteristics predict the magnitude of the perception gap between a user's public opinion estimate and EchoBreaker's estimate?

RQ2 (Response): When users are confronted with EchoBreaker's data that contradicts their own public opinion estimates, how do they respond? Does the response evolve with repeated exposures over a prolonged period of time?

RQ3 (Design): What features of a real-time, in situ calibration tool (like EchoBreaker) impact a user's receptivity to it, especially when it conflicts with their own perceptions?

Value to User Community: Users of EchoBreaker are anyone who uses social media who wants awareness on public opinion sentiment around various posts. My goal is that a tool like this closes perception gaps (especially regarding radical views) and helps people realize that we all have more in common than it may seem. Users can download this extension by taking the 'extension' file from the Github repository. I don't plan on launching this publicly on the Chrome extension store as EchoBreaker's purpose is as a research tool. However, my hope is that other researchers or entrepreneurs can build on top of my work; maybe someone else can launch something based on my learnings!

Demo:

I have already built the browser extension and the product has gone through multiple iterations based on user feedback (i.e. friends using it). It is now ready to be studied with 'real' participants with whom I have no prior relationship. I have already scheduled these user interviews.

You can see a very quick demo of EchoBreaker here:

<https://www.loom.com/share/7a0a692b99524de08819a7c9d833acda>

Note that this is NOT the video I will show for my actual demo presentation.

Delivery:

- I will include a github repository for EchoBreaker. This will be public so that anyone else can use it and build upon my work. It actually already exists.
<https://github.com/danielkyulee/echobreaker>
- I will present the findings/analysis of my pilot study (N = 2) that attempts to answer the 3 research questions above. To write my 'publishable research paper' (after this class), I plan to do a two-phase mixed-methods study. Phase 1 will be a longitudinal probe study (N = 10) where participants will use EchoBreaker for ~5 days. The results of this study will then help me form Phase 2 which will be a survey experiment (N = 200+). However, for the purposes of this class, I probably won't have time to finish either Phase 1 or Phase 2 as I want to get more feedback from HCI professors, figure out IRB approval, etc. Therefore, the N=2 pilot study is what I will present during my class presentation which I believe is well within the scope of this class; please let me know if I am wrong! I am treating that almost as 'my practice' before doing real studies and also as a chance to further work out bugs and UX issues with EchoBreaker.

Anything else?:

- I utilized Claude Code to develop Echobreaker.

Disclosure:

I have mentioned this to Dr. Kaiser already, but I am going to get feedback on this from Dr. Smith and Dr. Chilton (and maybe other professors) as I want to produce a paper for the HCI community. This EchoBreaker project/paper is not work I am doing for any other class. I guess it's possible that I will continue working on this after Spring as part of 6901 (or something like that) with Dr. Smith or Dr. Chilton, but that conversation has not occurred yet and may never occur.